

Score: / 10

Calculus I (MATH 2200)
Spring 2020
Wednesday, 02/26/2020
Quiz 5: Section 3.3, 3.4

Name (Print): _____

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Section 3.3

1. Find the indicated derivatives:

2 pts

(a) $\frac{d}{du}(2u^2 - 3u + 8)$

Using the sum rule and power rule, we find:

$$\frac{d}{du}(2u^2 - 3u + 8) = 4u - 3.$$

2 pts

(b) $f'(x)$, where $f(x) = x^{3.2} + 7x^{2.1} - 3x + 7$

Using the sum rule and power rule, we find:

$$\begin{aligned} f'(x) &= 3.2x^{3.2-1} + 7(2.1)x^{2.1-1} - 3 \\ &= 3.2x^{2.2} + 14.7x^{1.1} - 3. \end{aligned}$$

2 pts

(c) $f'(x)$, where $f(x) = x^{4/5} + 4x^{3/5} - 2x^{-1/5} + 5$

Using the sum rule and power rule, we find:

$$\begin{aligned} f'(x) &= (4/5)x^{4/5-1} + 4(3/5)x^{3/5-1} - 2(-1/5)x^{-1/5-1} \\ &= \frac{4}{5}x^{-1/5} + \frac{12}{5}x^{-2/5} + \frac{2}{5}x^{-6/5}. \end{aligned}$$

Section 3.4

4 pts

1. Let $f(x) = (4x^2 + 7)(2x^2 - 9)$. Find the value(s) of x for which the line tangent to the graph of f at $(x, f(x))$ is horizontal.

The slope of the line tangent to the graph of f at $(x, f(x))$ has slope $f'(x)$. So, using the product rule, we find $f'(x)$ as follows:

$$\begin{aligned} f'(x) &= \frac{d}{dx}(4x^2 + 7)(2x^2 - 9) + (4x^2 + 7) \frac{d}{dx}(2x^2 - 9) \\ &= (8x)(2x^2 - 9) + (4x^2 + 7)(4x) \\ &= 32x^3 - 44x \end{aligned}$$

Now, as a horizontal line has slope zero, we need only solve $f'(x) = 0$. So, let us note the following:

$$\begin{aligned} f'(x) &= 0 \\ 32x^3 - 44x &= 0 \\ x(32x^2 - 44) &= 0 \end{aligned}$$

So, we solve $f'(x) = 0$. From the above factorization, this occurs when $x = 0$, when $x = \sqrt{\frac{11}{8}}$, and when $x = -\sqrt{\frac{11}{8}}$. Correspondingly, the tangent line is horizontal for these values of x .

1 (bonus)

2. Find the first derivative of $f(x) = 3e^{2x}$.

First, use the properties of exponents to re-write $f(x) = 3e^{2x}$ as $f(x) = 3e^x e^x$. Now, using the product rule,

$$\begin{aligned} f'(x) &= \frac{d}{dx}(3e^x)(e^x) + (3e^x) \frac{d}{dx}(e^x) \\ &= (3e^x)(e^x) + (3e^x)(e^x) \\ &= 3e^{2x} + 3e^{2x} \\ &= 6e^{2x}. \end{aligned}$$